

PROGRAM:

```
#include <stdio.h>

#include <stdlib.h>

#define MAX 100 // Maximum size of the stack

// Stack structure definition

struct Stack {

    int arr[MAX];

    int top;

};

// Function prototypes

void initializeStack(struct Stack *stack);

void push(struct Stack *stack, int element);

int pop(struct Stack *stack);

void displayStack(struct Stack *stack);

int main() {

    struct Stack stack;

    int choice, element;

    initializeStack(&stack);

    while (1) {

        printf("\nStack Operations Menu:\n");

        printf("1. Push\n");

        printf("2. Pop\n");

        printf("3. Display\n");

        printf("4. Exit\n");

        printf("Enter your choice: ");

        scanf("%d", &choice);
```

```
switch (choice) {  
    case 1:  
        printf("Enter element to push: ");  
        scanf("%d", &element);  
        push(&stack, element);  
        break;  
    case 2:  
        pop(&stack);  
        break;  
    case 3:  
        displayStack(&stack);  
        break;  
    case 4:  
        exit(0);  
    default:  
        printf("Invalid choice! Please enter a valid option.\n");  
}  
}  
return 0;  
}  
  
// Function to initialize the stack  
void initializeStack(struct Stack *stack) {  
    stack->top = -1;  
}  
  
// Function to push an element onto the stack  
void push(struct Stack *stack, int element) {
```

```
if (stack->top == (MAX-1)) {  
    printf("Error: Stack overflow. Cannot push element.\n");  
    return;  
}  
stack->arr[++stack->top] = element;  
printf("Element pushed: %d\n", element);  
}  
  
// Function to pop an element from the stack  
int pop(struct Stack *stack) {  
    if (stack->top == -1) {  
        printf("Error: Stack underflow. Cannot pop element.\n");  
    }  
    else {  
        printf("\nPopped element: %d\n", stack->arr[stack->top--]);  
    }  
}  
  
// Function to display the stack elements  
void displayStack(struct Stack *stack) {  
    if (stack->top == -1) {  
        printf("Stack is empty.\n");  
        return;  
    }  
    printf("Stack elements: ");  
    for (int i = stack->top; i >= 0; i--) {  
        printf("%d ", stack->arr[i]);  
    }  
    printf("\n");  
}
```

}

OUTPUT:

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit Enter your choice: 1

Enter element to push: 10

Element pushed: 10

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit Enter your choice: 1

Enter element to push: 20

Element pushed: 20

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

Stack elements: 20 10

Stack Operations Menu:

1. Push
2. Pop

3. Display

4. Exit

Enter your choice: 2

Popped element: 20

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 3

Stack elements: 10

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 4